ASSOCIATED PRIMES OF A FIBRE OF A PROJECTIVE MODULE DO NOT MEET THE EXTENDED MAXIMAL IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The homothety spectrum and universal injectivity of the localisation away from the zerodivisors of the fibre. Same chapter flat.tex, labels lemma-homothety-spectrum and lemma-invert-universally-injective (proof lines 4331-4332); Algebra, definition algebra-definition-ord for the order used later. Gives the semi-locality statement and the universal injectivity of the map localising away from the elements that are not zerodivisors on the fibre, whose conclusions the proof uses.

Source reference. Original lemma 05DE: sources/stacks-flat/pinned-original-source.tex, source lines 1783–1783. Navigation pointer for \ref{lemma-homothety-spectrum} used inside the counted span (line 4331); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FR: sources/stacks-flat/pinned-original-source.tex, source lines 1976–1976. Navigation pointer for \ref{lemma-invert-universally-injective} used inside the counted span (line 4332); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a local ring with maximal ideal $\mathfrak{m}$. Let $R \to S$ be a ring map. Let N be an S-module. Assume*

(1) *N is projective as an R-module, and*
(2) *$S/\mathfrak{m}S$ is Noetherian and $N/\mathfrak{m}N$ is a finite $S/\mathfrak{m}S$-module.*

Then for any prime $\mathfrak{q} \subset S$ which is an associated prime of $N \otimes_R \kappa(\mathfrak{p})$ where $\mathfrak{p} = R \cap \mathfrak{q}$ we have $\mathfrak{q} + \mathfrak{m}S \neq S$.

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Proof. Note that the hypotheses of Lemmas 05DE and 05FR are satisfied. We will use the conclusions of these lemmas without further mention. Let $\Sigma \subset S$ be the multiplicative set of elements which are not zerodivisors on $N/\mathfrak{m}N$. The map $N \to \Sigma^{-1}N$ is R-universally injective. Hence we see that any $\mathfrak{q} \subset S$ which is an associated prime of $N \otimes_R \kappa(\mathfrak{p})$ is also an associated prime of $\Sigma^{-1}N \otimes_R \kappa(\mathfrak{p})$. Clearly this implies that $\mathfrak{q}$ corresponds to a prime of $\Sigma^{-1}S$. Thus $\mathfrak{q} \subset \mathfrak{q}'$ where $\mathfrak{q}'$ corresponds to an associated prime of $N/\mathfrak{m}N$ and we win. $\square$